

# MD SOAD SOLAIMAN

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[LinkedIn](#) | [GitHub](#)

## EDUCATION

### Master of Science in Data Science Jan 2026 – Present (Expected Nov 2026)

*The University of Adelaide, Adelaide, Australia*

GPA: 6.2 / 7.0

### Bachelor of Science in Industrial & Production Engineering Jan 2020 – Mar 2024

*Military Institute of Science and Technology (MIST), Dhaka, Bangladesh*

CGPA: 3.69 / 4.00

## RESEARCH & TECHNICAL PROJECTS (Available in GitHub)

### Trustworthy Foundation Models for Time-Series Forecasting *Independent Research, 2026*

- Developed a leakage-controlled, reproducible forecasting framework comparing Chronos, TimesFM, classical, LSTM, and naive models across Bitcoin and electricity datasets.
- Evaluated models using accuracy, statistical significance, robustness, temporal stability, uncertainty calibration, and multidimensional trustworthiness analysis.

<https://github.com/mdsoadsolaiman/Time-Series-Foundation-Models>

### MIT-BIH Arrhythmia Heartbeat Classification *Coursework Research Project, 2026*

- Built a patient-independent (record-level split) pipeline classifying four AAMI heartbeat classes from the MIT-BIH Arrhythmia Database, comparing Logistic Regression, Random Forest, and CNN models.
- Engineered 18 rhythm, morphology, and waveform features per heartbeat and applied class-balanced training to address severe class imbalance (majority class at 82.8% of beats).
- Evaluated models with per-class precision/recall/F1, precision-recall AUC, and McNemar's test for pairwise statistical comparison of minority-class recall.
- Related model complexity to minority-class performance and framed deployment trade-offs across screening, triage, and high-throughput clinical settings.

<https://github.com/mdsoadsolaiman/MIT-BIH-Heartbeat-Classification>

### Machine Learning & Deep Learning Portfolio *Projects 2026*

- Blood-Cell MiniResNet Classification: Built a MiniResNet-based pipeline for eight-class blood cell image classification, achieving 97.66% peak validation accuracy (96.88% accuracy from the saved checkpoint).
- Diabetes MLP Classification: Developed multilayer perceptron models for binary diabetes prediction from tabular clinical data, reaching a final test ROC-AUC of 0.8480.
- MNIST Classification and Clustering: Compared Logistic Regression and RBF-kernel SVM for digit classification (mean cross-validated F1 of 1.0000 and  $\approx 0.9699$  respectively) and applied PCA and K-Means for unsupervised digit-7 clustering.
- Butterfly CNN Classification: Designed a compact CNN for five-class butterfly image classification, reaching 72.22% accuracy on a held-out internal test set.

<https://github.com/mdsoadsolaiman/Machine-Learning-and-Deep-Learning-Projects>

### Advanced Analytics Portfolio *Projects 2026*

- Business Analytics — venue performance/revenue, player investment/compensation, and stadium profitability analyses.
- Public Data Analytics — the R/Quarto/Shiny app on South Australian rental affordability for international students.
- Statistical Research Analytics — the staged academic performance analysis (EDA → hypothesis testing → regression → diagnostics).

<https://github.com/mdsoadsolaiman/Advanced-Analytics>

### Executive BI Dashboard: Airline Workforce & Operations Analytics *Project 2026*

- Built a Tableau executive dashboard analysing cabin crew workforce performance, flight reliability, and operational decision-making for Qantas and Virgin Australia (2022 to 2025).

<https://github.com/mdsoadsolaiman/Business-Intelligence-Dashboard>

## PUBLICATIONS

Solaiman, M. S., Sakib, M. S. I., Rahman, M. A., & Rashid, A. B. (2024). Effect of cutting parameters on performance of CNC hot wire styrofoam cutting machine. December 2024.

DOI: 10.46254/BA07.20240050

Mahin, A. M. S., Solaiman, M. S., & Shahriar, B. M. (2024). Optimization of inbound logistics of automobile industry using FlexSim. December 2024.

DOI: 10.46254/BA07.20240083

## PROFESSIONAL EXPERIENCE

### Assistant Engineer, Business Operations & Development Feb 2024 – Dec 2024

*Anwar Group of Industries, Bangladesh*

- Supported business operations and process-development initiatives across manufacturing units, applying industrial engineering methods to operational problems.